

FAITH, SD, USA

WMO: 726539

Lat: 45.032N

Lon: 102.019W

Elev: 2580

StdP: 13.38

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 94056

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.4	-6.1	-17.1	2.4	-9.9	-11.9	3.2	-5.2	35.1	29.5	32.6	27.5	11.8	290	0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	96.7	67.5	92.2	66.8	88.8	66.0	72.2	85.8	70.4	84.6	68.8	82.9	12.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	114.2	77.8	66.1	105.8	76.1	64.2	98.8	74.6	37.6	86.0	35.9	84.5	34.5	82.5	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.7	28.4	25.4	-17.2	102.4	4.6	5.6	-20.5	106.4	-23.2	109.7	-25.8	112.8	-29.1	116.9		
			-17.6	75.3	4.5	1.8	-20.9	76.6	-23.5	77.7	-26.1	78.7	-29.4	80.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	20.8	20.4	33.8	44.6	54.6	65.5	74.0	71.1	62.1	47.1	34.2	22.6
	DBStd	21.68	14.04	14.14	13.41	10.33	8.81	7.36	6.31	6.76	9.04	9.82	11.70	13.47
	HDD50	4037	905	830	518	215	54	1	0	0	11	171	482	850
	HDD65	7598	1370	1249	966	612	338	81	6	22	159	558	923	1314
	CDD50	2596	0	1	17	53	196	466	745	653	374	82	8	1
	CDD65	684	0	0	0	1	15	96	286	210	72	4	0	0
	CDH74	8356	0	0	12	50	243	1150	3389	2415	1002	95	0	0
	CDH80	3772	0	0	0	9	60	428	1689	1144	416	26	0	0
(14) Wind	WSAvg	12.8	13.4	12.6	13.8	14.2	13.4	11.9	11.5	11.5	12.5	13.3	12.8	12.9
(15) Precipitation	PrecAvg	16.7	0.3	0.4	1.1	1.9	3.0	3.0	2.3	1.5	1.1	1.0	0.5	0.4
	PrecMax	27.7	0.7	1.5	4.4	4.6	9.7	6.0	5.2	3.7	3.6	5.0	1.9	0.9
	PrecMin	9.8	0.0	0.0	0.0	0.1	0.9	0.4	0.4	0.3	0.0	0.0	0.0	0.0
	PrecStd	3.8	0.2	0.4	1.0	1.3	2.2	1.6	1.3	0.9	1.0	1.1	0.5	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	59.6	75.1	81.1	86.5	96.6	104.4	99.9	96.7	83.5	69.1	56.9
		MCWB	42.5	45.9	51.9	56.2	63.4	68.7	68.6	67.0	64.8	58.8	50.8	41.9
	2%	DB	47.7	49.3	66.7	72.8	80.6	89.7	97.5	95.2	90.2	74.9	61.2	49.5
		MCWB	38.3	40.1	48.8	54.3	61.9	67.0	68.6	66.7	63.4	55.3	46.9	38.5
	5%	DB	42.4	43.3	60.2	67.1	75.4	84.4	93.0	90.8	84.6	68.8	55.6	44.4
		MCWB	35.4	36.6	46.1	51.1	59.5	65.8	67.8	66.6	62.3	52.4	43.7	35.6
	10%	DB	38.6	38.4	52.9	62.1	71.2	80.9	89.7	86.5	79.7	63.1	50.4	39.1
		MCWB	33.6	33.5	43.0	48.3	57.2	64.5	66.8	65.4	60.3	49.9	40.7	33.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	45.7	54.1	58.3	68.1	74.4	75.2	73.2	69.1	60.4	51.3	42.6
		MCDB	55.4	58.8	68.7	74.9	79.4	83.9	87.4	88.0	84.6	78.4	68.3	53.2
	2%	WB	39.0	40.3	50.0	54.9	64.5	70.7	72.9	70.8	66.6	57.1	47.4	39.1
		MCDB	46.9	48.5	64.7	71.0	75.3	83.0	86.5	85.7	81.9	71.5	60.5	48.2
	5%	WB	36.0	36.7	46.7	52.0	61.3	68.6	71.0	69.0	64.5	53.7	44.3	35.8
		MCDB	41.5	42.7	59.8	65.4	72.5	80.9	85.3	83.5	79.5	66.0	54.8	42.9
	10%	WB	33.8	33.9	43.1	49.2	58.7	66.4	69.4	67.4	62.2	50.7	41.3	33.5
		MCDB	38.4	38.4	52.6	60.1	68.8	77.9	84.4	81.3	76.5	61.7	49.8	39.5
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	18.0	20.6	22.4	22.0	23.1	25.7	25.5	25.1	21.7	20.1	16.7
		MCDBR	22.6	23.3	32.1	32.6	29.4	29.4	31.4	32.0	32.7	31.3	27.4	21.5
		MCWBR	15.0	15.1	17.9	17.3	14.6	12.3	10.5	11.5	12.7	15.5	15.5	13.7
	5% WB	MCDBR	21.7	21.7	31.0	30.7	26.5	26.0	26.4	27.3	28.8	28.8	26.9	21.2
		MCWBR	15.5	14.5	17.6	16.8	14.5	13.0	11.1	11.4	12.5	15.1	15.6	14.4
(40) Clear-Sky Solar Irradiance	taub		0.256	0.261	0.281	0.322	0.343	0.355	0.373	0.367	0.332	0.292	0.263	0.250
	taud		2.538	2.535	2.534	2.415	2.426	2.423	2.407	2.401	2.487	2.566	2.550	2.541
	Ebn at Noon		282	300	306	298	293	289	282	280	283	284	276	273
	Edh at Noon		20	24	28	35	36	36	36	35	30	23	20	18
(44) All-Sky Solar Radiation	RadAvg		575	899	1247	1587	1885	2154	2209	1866	1459	948	630	473
	RadStd		39	68	97	100	186	154	92	119	96	103	61	29

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page